Durga Prasad Oduri

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SUMMARY:

- **Data Scientist / Gen-AI Engineer with 11+ years of experience** in designing, developing, and deploying enterprise-grade AI/ML and Generative AI solutions.
- Expertise in **Machine Learning, Deep Learning, NLP, Generative AI, and Predictive Modeling** across diverse industries including telecom, finance, and healthcare.
- Proficient in **Python, R, Scala, and Java**, with deep knowledge of libraries such as **NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, HuggingFace, NLTK, spaCy, Gensim**.
- Proficient in **Time Series Modeling and Forecasting** using RNNs, LSTMs, MMM, ARIMA, and SARIMA, with expertise in statistical methodologies such as Hypothesis Testing, ANOVA, multivariate statistics, decision theory, descriptive statistics, probability, and linear algebra for advanced model development.
- Skilled in the **entire Data Science lifecycle**, including Data Extraction, Data Cleaning, Exploratory Data Analysis (EDA), Data Pre-Processing, Feature Engineering, Dimensionality Reduction, Algorithm Implementation, Back Testing, and Validation for handling structured, unstructured, and time-series data.
- Experienced in **Deep Learning and AI algorithms**, including Artificial Neural Networks (ANN), CNNs, and RNNs, with strong skills in hyperparameter tuning and model optimization using Python frameworks such as TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SciPy, and NLTK.
- Expertise in **MLOps and cloud platforms**, implementing Azure CI/CD pipelines for collaboration, model validation, monitoring, and governance; proficient in AWS (**EC2, S3, EMR, Lambda, CloudWatch, SageMaker**), Google Cloud (Vertex AI), and Azure services for deploying scalable ML solutions with Docker, Kubernetes, and Flask APIs.
- Adept at **Data Visualization and Business Intelligence**, leveraging Tableau, Power BI, Matplotlib, Seaborn, Plotly, and ggplot2 to deliver actionable insights; experienced with relational and non-relational databases such as MySQL, SQL, Oracle, and PostgreSQL for data analysis and reporting.
- Implemented **Agentic workflows** integrating **retrieval-augmented generation (RAG)**, **memory-based context management**, and **multi-agent collaboration** for complex enterprise tasks.
- Strong domain expertise in **healthcare and GenAI solutions**, leveraging AWS SageMaker and cognitive services (speech and voice recognition, text analytics) for building and deploying end-to-end ML pipelines, ensuring compliance, scalability, and alignment with business goals.
- Proficient in **Python, R, SQL, PySpark, and SAS**, with experience in data manipulation, data mining, text mining, sentiment analysis, brand reputation monitoring, and applications across computer vision, IoT, risk analytics, and predictive analytics; familiar with Git for CI/CD, R Studio, Spark Core, RDDs, DataFrames, MapReduce, and Spark-SQL for cloud-based ETL processes.
- Skilled in ML/LLM engineering, including model fine-tuning, Explainable AI (XAI), LoRA, RAG, PEFT, Knowledge Graphs, agentic programming, and **LLMs such as GPT, LLaMA, Claude, BERT, RoBERTa, and LangChain**, with deep experience in **Google Cloud LLMs, Google Vector Search, and Hybrid Search methodologies**.
- Proven ability in designing and deploying **ML pipelines and MLOps workflows** using AWS SageMaker, SageMaker Pipelines, Docker, Kubernetes, Airflow, MLflow, CI/CD, and model monitoring; expertise in **building agentic systems like GraphRAG, Chain of Thought (CoT), and Tree of Thought (ToT)**.
- Hands-on experience in **Deep Learning frameworks** (PyTorch, TensorFlow, Keras) for CNN, RNN, ANN, LSTM, GANs, Reinforcement Learning, and Transfer Learning, with applications in speech recognition, image processing, time series forecasting, autoencoders, and recommender systems.
- Extensive knowledge in **Big Data and Data Warehousing ecosystems**, including Hadoop (Hive, Pig), Spark, AWS EMR, ETL tools (Informatica), OLTP/OLAP systems, Star and Snowflake schemas, dimensional modeling, ER modeling, and schema design using Erwin.

- Strong background in **data analytics, mining, cleansing, migration, profiling, transformation, and visualization**, leveraging Tableau, Power BI, R Shiny, QuickSight, Kibana, and Python libraries (Matplotlib, Pandas, NumPy) to deliver actionable insights and business intelligence.
- Designed and trained **Generative AI models** (LLMs, VAEs, GANs, Diffusion Models) using **Python**, **PyTorch**, and **TensorFlow**
- Built strong relationships with community advocates and early adopters, empowering them to create tutorials, integrations, and resources that expanded the ecosystem around Claude.
- Proficient in **cloud platforms and distributed computing** with hands-on expertise across AWS (SageMaker, EC2, Lambda, EMR, S3, Athena, Aurora, Lex, Recognition, API Gateway, IAM, CloudFormation, CloudWatch, Glacier) and Google Cloud services (Vertex AI, BigQuery, Cloud Composer, Storage).
- Adept in **end-to-end Data Science lifecycle**, including data extraction, web scraping, statistical analysis (T-test, ANOVA, ROC/AUC, Precision/Recall), NLP (SpaCy, NLTK, Hugging Face Transformers), and large-scale structured/unstructured data processing, with proven leadership in **Agile/Scrum collaboration across distributed teams**.

TECHNICAL SKILLS:

- **Deep Learning :** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Autoencoders, Transfer Learning(VGG, ResNet, InceptionNet, MobileNet), Attention Mechanism, Object Detection (YOLO, Faster R-CNN), OCR, Image Segmentation (U-Net, Mask R-CNN), Activation Functions (ReLU, Sigmoid, Tanh, Softmax), Optimization Algorithms (SGD, Adam, RMSprop), Loss Functions (Cross-Entropy, MSE), Regularization (Dropout, Batch Normalization), Model Deployment, TensorFlow, Keras, PyTorch
- **Generative AI :** LM's, Ollama, Langchain, Langsmith, Agentic AI, Fine-tuning Techniques (LoRA, QLoRA), Local LLMs (Mistral, Llama, Gemma, etc.), Inference APIs (Groq, Nvidia NIM, etc.), Retrieval-Augmented Generation (RAG), Graph Databases(Neo4j), Vector Databases (Faiss, Chroma, Pinecone), Embedding Models(Mistral, Google AI Embedding Models), Prompt Engineering, Prompt Templates, Prompt Versioning, Document Chunking & Indexing, Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Conditional GANs, CycleGAN, Chatbots, AI Search Engines, Semi-Supervised Learning.
- **LLM & Frameworks:** Claude 3, LangChain, AutoGen, CrewAI, LlamaIndex, LangGraph
- **Natural Language Processing (NLP) :** Text Preprocessing (Tokenization, Lemmatization, Stemming), Bag of Words (BoW), TF-IDF, Word Embeddings (Word2Vec, GloVe, FastText), Sequence-to-Sequence Models, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), Transformer Models (BERT, GPT, T5), Named Entity Recognition (NER), Part-of-Speech Tagging, Sentiment Analysis, Hugging Face.
- **MLOps & LLMOps:** CI/CD for ML, Model Deployment (REST API, Batch), Model Monitoring, Model Versioning, Model Retraining Pipelines, ML Pipelines (MLflow, Airflow, Kubeflow), Experiment Tracking (MLflow, Weights & Biases), Data Versioning (DVC), Logging and Alerting (Prometheus, Grafana), Containerization (Docker), Orchestration (Kubernetes), Feature Stores (Feast), Cloud ML Platforms (AWS SageMaker, AWS BedRock, GCP AI Platform, Google Vertex AI, Azure AI) LLM Response Evaluation (TruLens, PromptLayer), Model Registry, Token Usage Tracking, Scalable LLM Deployment (TGI, vLLM).
- **Amazon Web Services :** EC2, Lambda, Sage Maker, Bedrock, EMR, S3, Glue, MKS, Kinesis, Quick Sight, API Gateway, Athena, Lex, Recognition, CI/CD, Code Commit, DynamoDB, transcribe, Cloud Formation, Cloud Watch, Glacier, IAM.
- **Database Servers :** MySQL, Mongo Microsoft SQL server, SQLite, Red Shift, RDS, PostgreSQL, SQLAlchemy, Mongo DB, Teradata.
- **Other Tools & Technologies :** Git, GitHub, GitLab, Docker, Docker Compose, Kubernetes, VS Code, Jupyter Notebook, Google Colab, CUDA Toolkit, Postman, Swagger, REST APIs, Linux/Unix Command Line, Bash Scripting, Conda, Virtualenv, Makefile, YAML, JSON, Terraform, Nginx, Redis, Jenkins, Power BI, Tableau, Excel, Google Sheets, Notion, Trello, Slack, Figma

EDUCATION:

- Bachelors of Engineering from Jawaharlal Nehru Technological University- 2012

EXPERIENCE:

Client: Wycliffe, Remote

Jan 2023 – Till Date

Role: Sr. Gen AI Engineer

Responsibilities:

- Engineered and deployed end-to-end Machine Learning solutions using Python, R, and SQL for predictive analytics, anomaly detection, and churn prediction, achieving measurable reductions in customer attrition.
- Applied advanced Natural Language Processing (NLP) techniques such as tokenization, lemmatization, sentiment analysis, topic modeling, and named entity recognition to extract insights from large-scale unstructured datasets.
- Designed, trained, and optimized Deep Learning models (CNNs, RNNs, LSTMs, Transformers) using TensorFlow, Keras, and PyTorch, improving recommendation accuracy and classification performance.
- Designed and implemented domain-adapted Generative AI solutions using advanced LLM fine-tuning and model distillation techniques, enabling efficient deployment of large models for high-throughput, enterprise-scale recommender and decision-support systems.
- Built production-grade GenAI pipelines incorporating prompt engineering, embeddings, and feedback-driven optimization, leveraging developer telemetry and AI coding assistant usage to continuously improve model accuracy, relevance, and usability.
- Architected and governed secure, compliant, and traceable agentic AI workflows, integrating human-in-the-loop controls and auditability across complex LLM- and agent-based systems to meet enterprise security, privacy, and ethical AI standards.
- Designed and deployed multi-agent systems using CrewAI, orchestrating role-based agents (Planner, Researcher, Executor) for autonomous document analysis and decision support.
- Architected stateful, graph-based agent workflows with LangGraph, enabling deterministic control over multi-step reasoning paths and complex business logic.
- Integrated AI agents with enterprise data sources and external APIs, allowing agents to retrieve real-time data and take contextual actions within secure environments.
- Led RAG pipeline design utilizing vector databases (Pinecone, FAISS) to improve LLM inference latency and response quality for high-throughput client-facing systems.
- Governed secure AI agent workflows by integrating human-in-the-loop controls and auditability, ensuring 100% alignment with enterprise privacy and ethical AI standards.
- Developed scalable conversational AI solutions, fine-tuning models to interpret user intent and automate customer support with high precision.
- Researched, evaluated, and piloted cutting-edge LLMs and multimodal generative models, driving adoption of Azure OpenAI and Microsoft Copilot Studio to accelerate AI-driven automation and intelligent decision-making initiatives at NatWest.
- Led prompt engineering strategy and RAG pipeline design, implementing optimized data chunking, tokenization, and retrieval strategies to improve LLM inference latency, contextual grounding, and response quality in production environments.
- Developed scalable cloud-native GenAI architectures using GCP Vertex AI and Databricks ML Runtime, performing feature engineering, model training, and hyperparameter optimization with Hyperopt, and monitoring real-time performance, latency, and cost via Streamlit-based dashboards.
- Delivered LLM-powered conversational AI solutions, fine-tuning NLP and generative models to accurately interpret and classify user intent, significantly improving chatbot precision and enabling intelligent customer support automation.

- Implemented LLM fine-tuning using LoRA, PEFT, and full-parameter training, optimizing contextual accuracy and cost efficiency for enterprise Generative AI use cases while benchmarking performance across multiple model variants.
- Collaborated closely with cross-functional engineering, data, and business teams, mentoring junior engineers and contributing to internal knowledge-sharing initiatives on emerging GenAI, agentic frameworks, and best practices.
- Integrated AI agents into enterprise workflows using modern frameworks and orchestration patterns, enabling autonomous task execution, decision support, and scalable AI-driven automation aligned with strategic business objectives.
- Designed and deployed production-grade Generative AI systems using agentic frameworks, RAG pipelines, and multimodal LLMs, enabling scalable AI-driven automation and intelligent decision-making across enterprise workflows.

  Designed and implemented production-grade LLM applications using LangChain, including RAG pipelines, tool calling, prompt templates, memory management, and multi-step reasoning chains.
- Built end-to-end agentic workflows integrating vector databases (FAISS, Pinecone, Weaviate) with LangChain retrievers to enable low-latency, context-aware document Q&A and enterprise search systems.
- Developed reusable prompt engineering and chain abstractions, improving consistency, maintainability, and scalability across multiple GenAI applications.
- Architected and deployed multi-agent systems using CrewAI, enabling autonomous task execution through role-based agents such as planner, researcher, executor, and reviewer.
- Implemented collaborative agent workflows for enterprise use cases including document analysis, compliance validation, workflow automation, and decision support systems.
- Integrated CrewAI agents with external tools, APIs, and enterprise data sources, enabling agents to retrieve data, reason over results, and take contextual actions.
- Integrated LangChain with Azure OpenAI, AWS Bedrock, and GCP Vertex AI, supporting multi-model orchestration and fallback strategies for high-availability production deployments.
- Built and operationalized LLMOps pipelines covering prompt engineering, embeddings, fine-tuning (LoRA/PEFT), evaluation, and monitoring, optimizing model performance, latency, and cost in real-world production environments.
- Architected cloud-native GenAI solutions across Azure OpenAI, AWS Bedrock, and GCP Vertex AI, integrating secure APIs, model orchestration, and scalable inference patterns for high-availability deployments.
- Implemented secure and compliant AI governance frameworks, including human-in-the-loop workflows, auditability, and responsible AI controls to meet enterprise security, privacy, and ethical AI standards.
- Led cross-functional collaboration and technical mentorship, translating complex GenAI and agentic AI concepts into actionable solutions that aligned with business strategy and accelerated enterprise AI adoption.
- Conducted in-depth data analysis with Python, R, and SQL, extracting actionable insights to drive data-driven decisions across NatWest's business units.
- Led high-impact educational initiatives—office hours, Q&A sessions, workshops, and prompt-crafting tutorials—to help members use Claude and similar AI tools effectively and responsibly.
- Collaborated with cross-functional teams to implement community-driven improvements, closing 200+ Jira tickets tied to UX enhancements and documentation updates.
- Designed and managed project workflows in Trello, improving team efficiency and ensuring timely execution of community campaigns.
- Created Figma-based educational materials and step-by-step guides.
- Collaborated with AI engineers to design user journeys in Figma for beta testing new tools, resulting in actionable feedback and UX improvements.
- Collaborated closely with product and engineering teams to translate community feedback into actionable insights, directly influencing updates to Gemini's UI/UX and model workflows.

- Logged, tracked, and prioritized community-reported issues, feature requests, and bug reports in Jira, ensuring smooth handoffs to product and engineering teams.
- Developed MapReduce/Spark Python modules for machine learning and predictive analytics in Hadoop on AWS and HPE's cloud platforms (EMR, S3, Apache Spark), ensuring scalability and high-performance.
- Developed and optimized ML and AI models using TensorFlow, PyTorch, and HPE AI/ML tools, focusing on computer vision, natural language processing (NLP), and predictive analytics.
- Automated document understanding and workflow generation using RAG + Agentic orchestration pipelines with knowledge graphs.
- Coordinated cross-functional teams using Trello boards, automating task assignments and tracking milestones for faster project delivery.
- Launched new community platforms using Figma wireframes to guide onboarding experiences, increasing member activation rate
- Managed the full lifecycle of community feedback by logging bugs, suggestions, and user pain points in Jira, improving engineering response time
- Partnered with cross-functional teams to resolve Jira-tracked issues, contributing to faster bug fixes and improved user satisfaction scores
- Nurtured a vibrant AI community of creators, developers, and everyday users by hosting discussions, live demos, and learning sessions centered on Claude, ensuring members felt supported, heard, and inspired.
- Collaborated cross-functionally with engineering, marketing, and support teams by using Copilot to structure reports, extract themes from large data sets, and prepare stakeholder briefs.
- Specialized in deep learning solutions such as image classification, object detection, and language translation, applying hyperparameter tuning, model pruning, and transfer learning to maximize performance.
- Implemented regression models (Linear, Ridge, Lasso) and classification models (Logistic Regression, SVM, Random Forest, GBM, XGBoost) to predict financial outcomes and customer behaviors.
- Engineered ensemble models (Bagging, Random Forest, XGBoost, AdaBoost), improving accuracy by 40%, reducing bias/variance by 30%, and boosting operational efficiency by 50%.
- Created and deployed recommendation systems leveraging content-based, collaborative filtering, ANN deep learning, and LightFM, increasing recommendation precision by 60%, CTR by 35%, and user satisfaction by 25%.
- Applied clustering algorithms (K-means, Gaussian mixture, DBSCAN) for customer segmentation, product pricing optimization, and targeted marketing campaigns at NatWest.
- Designed and implemented RAG pipelines using vector databases (Pinecone, Weaviate, FAISS) for document retrieval and context-based query resolution in compliance and customer service use cases.
- Integrated RAG workflows with LLMs to enhance document summarization, Q&A systems, and personalized customer interactions.
- Mentored junior community managers, creating onboarding guides and templates in Trello to standardize processes and best practices.
- Coordinated beta testing groups for new AI features, capturing structured feedback, logging test results in Jira, and presenting insights to improve release quality.
- Transformed user feedback into clear insights for product teams, helping refine Claude's features, reduce friction points, and strengthen the overall user experience.
- Produced visual guides and templates for social media content using Figma, improving team efficiency and brand consistency.
- Authored clear, friendly, and humanized communication materials—including updates, tutorials, onboarding guides, and release notes—to help users adopt and understand Gemini's evolving capabilities.
- Built a scalable community operations workflow by integrating Copilot for content ideation, moderation queues, and meeting recap summaries, improving team efficiency

- Collaborated closely with AI educators and power users to create workshops and onboarding materials that helped newcomers understand Claude's capabilities and responsible usage.
- Coordinated beta-testing programs for AI features, documenting feedback in Jira and influencing product iterations based on real-world use.
- Designed Figma-based onboarding flows and visual dashboards that improved new member activation
- Implemented secure AI governance and human-in-the-loop workflows ensuring traceability and compliance in Agentic pipelines.
- Used Microsoft Copilot to draft high-quality community updates, discussion prompts, and help-center content, enabling faster publishing while maintaining an authentic, human tone.
- Designed and hosted community events such as AMAs, workshops, and challenge weeks that boosted user participation and showcased creative Gemini use cases.
- Researched and experimented with Large Language Models (GPT) and explored Azure Microsoft Copilot Studio for generative AI adoption at NatWest.
- Guided community members on how to use Microsoft Copilot effectively through live demos, short-form tutorials, and user-friendly walkthroughs that boosted tool adoption.
- Developed and fine-tuned NLP models to interpret and categorize user queries, improving chatbot response accuracy and enabling smarter customer support automation.
- Implemented fine-tuning of LLMs using LoRA, PEFT, and full model training, boosting contextual accuracy in NatWest's Generative AI solutions.
- Validated AI data models using K-fold cross-validation to mitigate overfitting/underfitting in deep learning workflows.
- Developed and maintained Figma templates for tutorial creation, knowledge base guides, and event assets, improving internal workflow efficiency.
- Built trust with power users and early adopters by providing transparent updates, proactive communication, and personalized follow-ups on Jira-tracked issues.
- Leveraged analytics tools to track sentiment, trending AI topics, and member needs, using insights to shape monthly community strategy and highlight areas where Claude could be improved.
- Designed and delivered interactive learning sessions that taught members how to use Copilot and other AI productivity tools, resulting in higher self-service adoption.
- Deployed AI/ML algorithms on edge devices using HPE's platforms, supporting IoT and industrial automation scenarios.
- Built scalable microservices and APIs to deliver LLM-powered solutions in real-time NatWest applications.
- Developed and maintained a Data Dictionary and metadata reports for technical and business consumption.
- Created dynamic and interactive dashboards with Power BI and Tableau, presenting insights to stakeholders and enabling informed decision-making in risk, compliance, and customer analytics.
- Ensured all AI models complied with ethical AI principles, emphasizing fairness, transparency, and accountability, and aligning with NatWest's regulatory standards.
- Collaborated with cross-functional teams using Git for version control and implemented CI/CD pipelines for seamless ML model deployment.
- Designed interactive Figma wireframes for new community platforms, improving user onboarding and engagement.
- Built and maintained Trello boards to manage community projects, track progress, and ensure on-time delivery of campaigns and events.
- Introduced AI-driven moderation workflows using Gemini for flagging harmful content, helping keep the community safe while reducing manual workload for moderators.
- Actively participated in Agile ceremonies including sprint planning, daily standups, backlog grooming, and client discussions, ensuring alignment with project timelines and business goals.

Client: ServiceNow Inc, Santa Clara, CA **Apr 2018 - Dec 2022**
Role: Data Scientist
Responsibilities.

- Designed and deployed multi-layered data pipelines to seamlessly transform raw data into actionable insights, enabling efficient workflows in Generative AI and LLM applications.
- Utilized Microsoft Copilot to draft community announcements, pre-screen FAQs, and generate fast, empathetic responses—boosting overall engagement and reducing manual workload.
- Built and optimized data lakehouse architectures to unify the strengths of data lakes and data warehouses, enhancing data governance, accessibility, and storage efficiency.
- Implemented structured moderation systems using both manual techniques and AI-assisted tools (including Gemini), reducing response time to community issues
- Managed the full lifecycle of community feedback by logging bugs, suggestions, and user pain points in Jira, improving engineering response time
- Coordinated cross-functional teams via Trello, streamlining task delegation, deadlines, and workflow visibility for smoother collaboration.
- Developed agent-to-agent conversational systems using AutoGen, facilitating dynamic collaboration for complex problem-solving and data analysis tasks.
- Optimized agent execution paths to significantly reduce token usage and latency while maintaining high accuracy in automated task completion.
- Built human-in-the-loop workflows, allowing for seamless escalation and review within autonomous agent pipelines to meet governance standards.
- Managed full-lifecycle data pipelines for LLM applications, transforming raw data into high-quality training corpora for fine-tuning and RAG integration.
- Implemented real-time monitoring and alerting for AI pipelines using Prometheus and Grafana to ensure system resilience and performance.
- Developed agent-to-agent conversational systems using AutoGen, enabling dynamic collaboration between LLM-powered agents for complex reasoning and problem-solving tasks.
- Built human-in-the-loop agent workflows, allowing seamless escalation, review, and approval within autonomous agent pipelines to meet enterprise governance standards.
- Integrated CrewAI agents with external tools, APIs, and enterprise data sources, enabling agents to retrieve data, reason over results, and take contextual actions.
- Designed agent memory and inter-agent communication patterns, improving task accuracy, traceability, and overall system reliability.
- Optimized agent execution paths to reduce latency and token usage while maintaining high task completion accuracy.
- Implemented tool-augmented AutoGen agents capable of code execution, API interaction, and knowledge retrieval to support data analysis, automation, and engineering use cases.
- Applied guardrails, safety policies, and conversation constraints to ensure secure, compliant, and predictable agent behavior in production environments.
- Designed stateful, graph-based agent workflows using LangGraph, enabling deterministic control over multi-step LLM reasoning and decision paths.
- Modeled complex business processes as directed graphs, supporting branching, looping, and conditional execution for robust enterprise automation.
- Integrated LangGraph with RAG pipelines, memory stores, and tool execution layers, ensuring consistent state management across long-running agent tasks.
- Implemented failure handling, retries, and checkpointing within LangGraph flows to improve reliability and observability of agentic AI systems.
- Implemented data cleaning and preprocessing frameworks that improved dataset quality, ensuring relevance, consistency, and accuracy for large-scale LLM training and NLP tasks.

- Automated repetitive community tasks—such as meeting notes, FAQ creation, and feedback summaries—through Copilot workflows, freeing time for deeper user engagement.
- Transformed user feedback into clear insights for product teams, helping refine Claude's features, reduce friction points, and strengthen the overall user experience.
- Developed content calendars, campaign strategies, and Trello templates, standardizing processes and improving team efficiency.
- Conducted comprehensive analyses of embedding techniques (word embeddings, contextual embeddings, vector embeddings), improving semantic search, data representation, and retrieval performance in Generative AI systems.
- Developed scalable data processing workflows integrating multiple structured and unstructured data sources, providing comprehensive coverage for LLM fine-tuning and RAG pipelines.
- Established a data quality assessment protocol ensuring datasets adhered to high accuracy, completeness, and integrity standards prior to machine learning integration.
- Partnered with customer success and product teams to translate community conversations into actionable Copilot prompt libraries, improving user onboarding and retention.
- Launched educational campaigns featuring Gemini's generative features (prompting best practices, model updates, demo showcases), driving a noticeable growth in user-generated content.
- Automated repetitive data ingestion and ETL processes using Python, Pandas, and PySpark, significantly improving efficiency and reducing errors in preparing datasets for Retrieval-Augmented Generation (RAG) systems.
- Created and managed vector databases/vector stores with optimized retrieval techniques (FAISS, Pinecone, Weaviate), enabling low-latency access to critical knowledge bases for real-time AI inference.
- Designed data cleaning and enrichment pipelines capable of handling heterogeneous data formats (structured, semi-structured, unstructured), ensuring high-quality inputs for LLMs and NLP frameworks.
- Built an automated data preparation framework, accelerating the integration of new datasets into training pipelines, RAG workflows, and model fine-tuning processes.
- Applied Reinforcement Learning with Human Feedback (RLHF) to improve LLM behavior, leveraging iterative evaluation loops and feedback signals to enhance generative response quality.
- Leveraged Dask and distributed computing frameworks to implement parallelized workflows, reducing latency, training time, and pipeline execution costs.
- Enhanced LLM user interaction through similarity search and semantic retrieval algorithms, improving response accuracy and contextual relevance.
- Developed content calendars, Trello templates, and process guides to standardize workflows and boost team productivity.
- Launched community feedback programs enhanced by Copilot auto-summarization, reducing review time and speeding up delivery of insights to engineering teams.
- Championed the adoption of Gemini across community channels by creating relatable, real-world demos and walkthroughs that helped users understand complex AI features with ease.
- Engineered robust data ingestion protocols for diverse data types (JSON, Parquet, CSV, XML, APIs), ensuring scalability and reliability in AI-driven applications.
- Utilized cloud-native solutions (AWS, GCP, Azure) for data storage, orchestration, and scalability, ensuring availability and performance of large-scale LLM training datasets.
- Developed state-of-the-art data processing workflows incorporating the latest LLM, RAG, and generative AI advancements to maximize system performance.
- Implemented real-time monitoring and alerting systems for ETL/data pipelines, enabling proactive detection and resolution of data quality issues.
- Coordinated cross-functional teams via Trello, streamlining task delegation, deadlines, and workflow visibility for smoother collaboration.
- Analyzed community sentiment using Gemini-driven insights and dashboards, identifying trends early and helping leadership make data-backed decisions.
- Served as the "voice of the community" during sprint planning, advocating for user needs, clarifying Jira tickets, and aligning product priorities with user feedback.

- Trello Trello
- Cultivated ambassador and beta-tester programs that empowered key community members to co-create with Gemini, resulting in a richer, more collaborative environment and accelerated product feedback cycles.
- Authored comprehensive technical documentation for data pipelines and workflows, improving collaboration across cross-functional data science and engineering teams.
- Established a data quality lifecycle management framework, continuously monitoring and optimizing dataset reliability across MLOps pipelines.
- Improved retrieval accuracy in RAG frameworks through iterative testing, evaluation, and fine-tuning of vector search methods.
- Collaborated with machine learning engineers, data engineers, and business stakeholders to optimize workflows, ensuring alignment of data solutions with business KPIs.
- Designed and deployed data lakehouse solutions with advanced analytics integration, enabling predictive modeling, descriptive analytics, and executive decision-making.
- Conducted experiments on vector search methodologies (ANN, cosine similarity, semantic embeddings) to benchmark performance improvements in retrieval-augmented AI systems.
- Developed protocols for dataset chunking and tokenization strategies, improving efficiency in LLM pre-training and inference pipelines.
- Created and managed Trello boards to streamline project planning, task delegation, and milestone tracking across multiple teams.
- Championed the adoption of Gemini across community channels by creating relatable, real-world demos and walkthroughs that helped users understand complex AI features with ease.
- Optimized data ingestion workflows for diverse sources, ensuring flexibility, scalability, and resilience in AI/ML ecosystems.
- Enhanced RAG pipeline performance via continuous retrieval optimization, prompt engineering, and evaluation metrics (BLEU, ROUGE, perplexity).
- Integrated Snowflake with enterprise systems, enabling seamless data sharing, governance, and federated analytics within cloud-native AI pipelines.
- Designed and implemented automated data refinement systems, improving the quality of training corpora for NLP/LLM models.
- Streamlined onboarding for new members by developing a clear knowledge base and Gemini-powered help center, decreasing new-user confusion and support tickets.
- Designed and managed project workflows in Trello, improving team efficiency and ensuring timely execution of community campaigns.
- Used JSON, REST APIs, and HTTP protocols to integrate heterogeneous data sources into data pipelines and ML workflows, enhancing system interoperability.
- Conducted performance and reliability assessments of cloud-based storage solutions, ensuring optimal scalability, security, and cost efficiency for AI-driven applications.

Client: USAA, San Antonio, TX **July 2017 - Mar 2018**
Role: Data Scientist
Responsibilities:

- Partnered with data engineers and operations teams to implement ETL processes; wrote and optimized complex SQL queries for financial data extraction, boosting query performance by 30%.
- Designed and built database models, APIs, and Views using Python and Flask, enabling interactive web-based analytical solutions adopted by 3+ business units.
- Conducted univariate and multivariate statistical analysis to uncover hidden patterns, correlations, and risk factors across credit, fraud, and financial transaction datasets.
- Leveraged Python (Pandas, NumPy, Scikit-learn, SciPy, Seaborn, Matplotlib, NLTK) to develop AI/ML models, including XGBoost, Random Forest, and SVM, reducing loan default risks by 18%.

- Developed predictive AI models using XGBoost and Random Forest, reducing financial risk factors and improving decision-making speed by 20%.
- Automated ML model deployment via AWS SageMaker and Docker, reducing manual intervention by 40%.
- Conducted NLP on unstructured financial documents to uncover insights, increasing compliance efficiency by 25%.
- Designed Snowflake schema data models and scalable ETL pipelines, ensuring financial datasets were ML-ready with 100% data lineage and audit compliance.
- Automated ML model deployment pipelines with Docker and deployed models on AWS SageMaker, reducing manual intervention by 40%.
- Implemented hybrid search strategies by integrating SQL-based structured queries with semantic NLP models, enhancing fraud detection accuracy by 22%.
- Applied data imputation and feature engineering using Scikit-learn, ensuring models maintained high accuracy despite incomplete or inconsistent data.
- Researched and implemented IoT-driven financial analytics solutions, integrating device-generated transactional data for real-time fraud monitoring and portfolio optimization.
- Applied strong analytical skills and critical thinking to solve complex financial challenges, improving model interpretability and business decision-making.
- Developed content calendars, campaign ideas, and engagement challenges through Copilot brainstorming workflows, resulting in more dynamic, user-led participation.
- Conducted text mining and Natural Language Processing (NLP) on unstructured financial documents, regulatory filings, and customer feedback, uncovering insights that increased compliance efficiency by 25%.
- Built Named-Entity Recognition (NER) models to automatically extract entities such as customer names, accounts, organizations, and locations from unstructured text, reducing manual review efforts by 40%.
- Applied clustering algorithms (K-Means, DBSCAN, Hierarchical) for client segmentation and anomaly detection in large-scale financial datasets, conducted geospatial analysis on ATM/branch transaction data to identify fraud hotspots, and partnered with stakeholders for feasibility analysis and requirements gathering, achieving 18% improved fraud detection, 12% cost savings, and up to 60% reduction in project expenses.
- Designed and trained deep learning models using TensorFlow and PyTorch to enhance predictive accuracy of loan approval and credit risk scoring models by 22%, while building scalable data science workflows in Databricks leveraging Apache Spark for distributed processing that reduced pipeline execution time by 45%.
- Engineered end-to-end data engineering pipelines leveraging AWS (S3, RDS) and Snowflake, automated robust ETL and orchestration workflows with Airflow and Databricks, and ensured 99.9% data availability for ML models while reducing pipeline latency by 35%.
- Designed and optimized mathematical/statistical models for financial forecasting, leveraging core computer science and algorithms principles for scalable solutions.
- Applied machine learning (ML) algorithms for risk modeling, portfolio optimization, and real-time fraud detection, reducing false positives by 20%.
- Designed and deployed text classification and sentiment analysis models for unstructured and semi-structured financial data, achieving a 15% reduction in false positives.
- Built SEO-optimized predictive analytics solutions, enabling marketing and client engagement strategies that boosted digital presence by 70%.
- Partnered with analytics teams to develop time-series forecasting and optimization models, improving financial trend predictions by 20%.
- Engineered a multilingual E-Learning Management System (SCORM compliant) using Articulate 360 and Redwood tools, enhancing compliance training completion rates by 50%.
- Designed logical and physical data models with Star and Snowflake schema techniques in Erwin, supporting both Data Warehouse and Data Mart structures.
- Utilized Power Query in Power BI for data cleansing, pivoting/un-pivoting, and massaging large-scale financial datasets for visualization.

- Delivered ad-hoc reports for executives by writing advanced SQL queries, reducing reporting turnaround time by 30%.
- Performed data validation, verification, cleansing, and quality checks to resolve mismatches and ensure integrity across millions of transactional records.
- Designed and optimized OLTP and OLAP data models, applying normalization/de-normalization concepts to balance storage and performance needs.
- Created custom SQL scripts and reporting tools for recurring analytical requirements, saving the team 15+ hours per week in manual work.
- Built custom dashboards and reports in Power BI, enabling real-time insights into credit risk, fraud detection, and financial KPIs, which improved decision-making speed by 20%.

Xoriant Solutions Pvt. Ltd, India
Sr. Data Analyst **Feb 2013 - Nov 2016**
Responsibilities:

- Acquired, extracted, and integrated data from diverse primary and secondary sources (EHR, claims, operational, and financial data) into centralized databases/data warehouses for analysis.
- Designed and maintained data pipelines, ETL workflows, and SQL queries to automate ingestion, transformation, and validation of healthcare datasets.
- Performed data cleaning, preprocessing, and imputation of missing values to ensure data accuracy, completeness, and reliability across multiple clinical and business systems.
- Identified, monitored, and resolved data quality issues by removing duplicates, standardizing formats, and ensuring compliance with HIPAA data governance standards.
- Built interactive dashboards and pivot tables in Excel, Tableau, and Power BI, providing actionable insights for executive leadership and clinical teams.
- Conducted trend and variance analysis on healthcare product/service lines to optimize operations, control costs, and recommend discontinuation of underperforming offerings.
- Applied statistical methods, regression analysis, and hypothesis testing to support decision-making in patient care and operational strategies.
- Leveraged Python (Pandas, NumPy, Matplotlib, Seaborn) for data analysis and developed predictive models to forecast patient volumes, resource utilization, and revenue outcomes.
- Developed and implemented a KNN classification model to optimize product recommendations, resulting in improved sales performance and measurable profitability.
- Applied computer vision techniques (OpenCV) to automate quality control by detecting defects in product packaging; trained deep learning models in PyTorch to classify defective vs. non-defective items with >90% accuracy.
- Partnered with cross-functional management teams to prioritize business requirements, align reporting strategies, and deliver insights that drove key clinical and financial decisions.
- Generated and automated monthly and quarterly sales/revenue comparison reports, highlighting growth trends and providing leadership with actionable recommendations for business improvements.
- Designed and deployed a diagnostic survey tool to measure and predict team performance, enabling HR and operations to enhance workforce productivity.
- Created SQL stored procedures, views, and queries to support ad-hoc reporting requests for clinicians, finance, and compliance teams.
- Compiled and analyzed patient satisfaction, operational KPIs, and financial metrics, delivering insights to improve care quality and business efficiency.
- Built machine learning pipelines using scikit-learn and PyTorch, trained models on structured/unstructured healthcare datasets, and evaluated performance metrics (precision, recall, AUC).
- Produced summary statistics, cohort analyses, and segmentation reports to provide healthcare leaders with data-driven strategies for patient outcomes and service optimization.

- Delivered regular dashboards, scorecards, and reports on active projects and accounts, ensuring transparency and accountability for stakeholders.
- Collaborated with IT and data engineering teams to implement data governance best practices, metadata management, and role-based access controls for compliance.
- Documented processes created data dictionaries, workflows, and technical specifications, and maintained knowledge repositories to support ongoing analytics initiatives.